

POWER BI MODULE END PROJECT - DOCUMENTATION

Module-End

Power BI Module End Project – Logistics &
Transportation

Project Title

Fleet Performance & Delivery Efficiency Data
Analysis using Power BI

Submitted By

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ANB18 Batch

Project Overview :

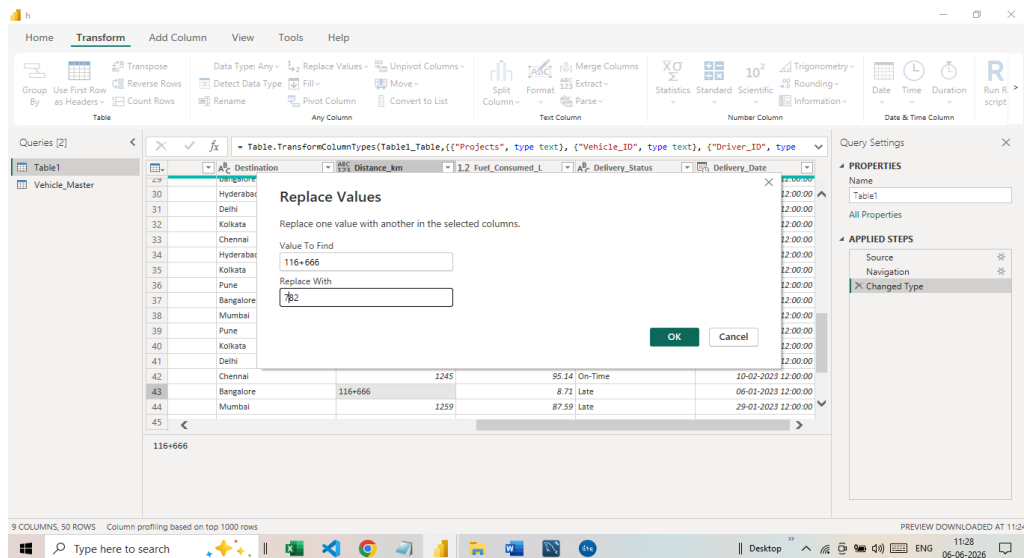
- ✓ This Power BI dashboard provides a centralized view of fleet operations across multiple routes and vehicle types in India.
- ✓ It tracks key performance indicators — including on-time delivery rate, fuel efficiency, cost per kilometer, and total trips — enabling operations teams to monitor and improve logistics performance.
- ✓ The dashboard supports dynamic filtering by vehicle type, origin, delivery status, destination, month, and driver ID, allowing stakeholders to drill into specific segments of the fleet.

Objectives :

- ✓ Monitor Delivery Performance: Track on-time delivery rates across all routes to identify bottlenecks and areas for improvement.
- ✓ Analyze Fuel Efficiency: Evaluate fuel consumption trends by vehicle and month to support cost reduction efforts.
- ✓ Track Operational Costs: Measure cost per kilometer across the fleet to assist in budget planning and cost optimization.
- ✓ Assess Vehicle Performance: Compare trip counts, fuel efficiency, and cost metrics per vehicle to flag underperforming assets.
- ✓ Visualize Geographic Coverage: Map delivery origins and destinations to understand route distribution and coverage across regions.
- ✓ Enable Data-Driven Decisions: Provide management with interactive, filterable insights to support strategic and operational decisions.

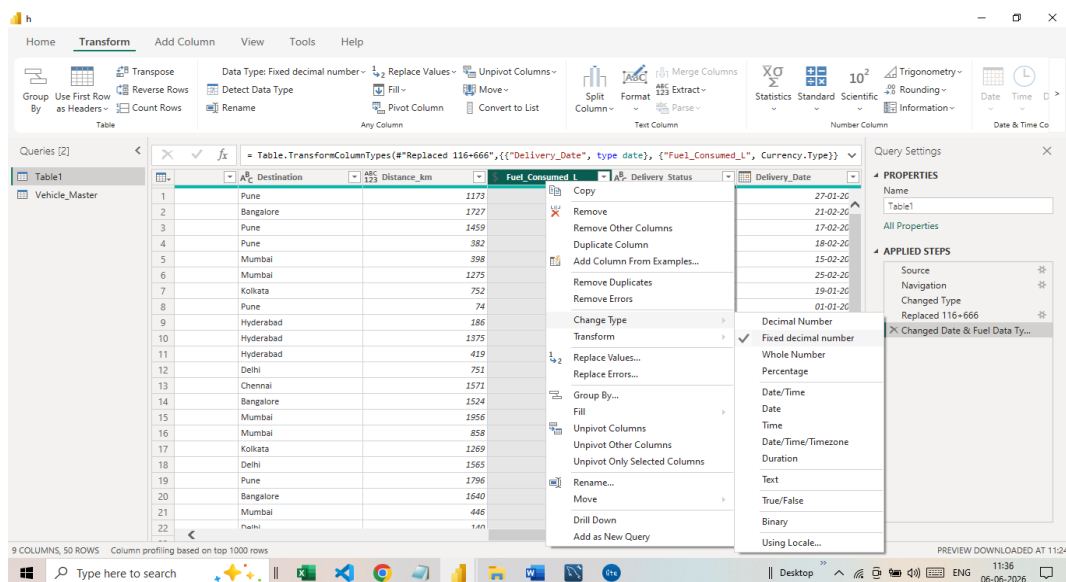
Power Query :

Replace Value



- ❖ As the first step of data cleaning in Power Query, I identified a null value in the 'Distance (km)' column, which was not relevant to the dataset. I corrected it using the 'Replace Values' option.

Data Type Change

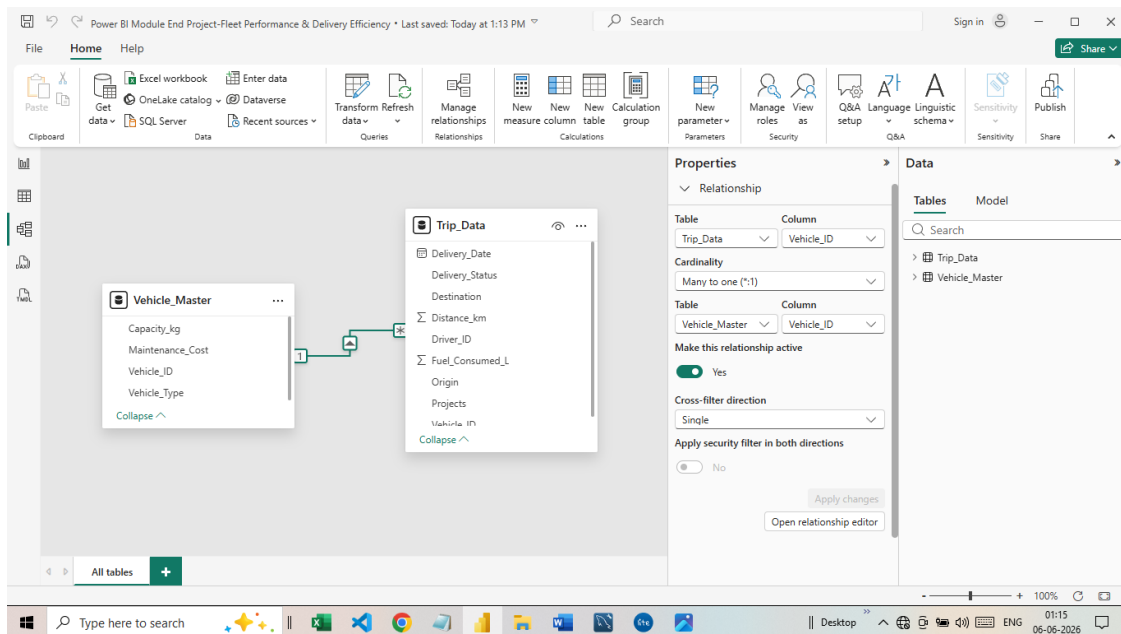


- ❖ Subsequently, Power Query automatically assigned data types to the columns, including the Date column. After reviewing the dataset, I modified the data

types of specific columns to align with my requirements, converting them to Fixed Decimal Number, Date, and Currency formats as appropriate.

- ❖ I converted specific text fields to uppercase, lowercase, or proper case formats, as required, to ensure consistency and improve data quality.

Data Modeling



- ❖ The dataset provided consisted of two tables: 'Trip Data' and 'Vehicle Master'. I established a relationship between them using the 'Vehicle ID' column, creating a Many-to-One relationship from Trip Data to Vehicle Master. The cross-filter direction was set to Single, and the relationship was configured as Active.

DAX

- ❖ This formula was extremely useful in creating the various KPI cards required for the dashboard. I have attached a screenshot of the formula below for reference.

Power BI Module End Project-Fleet Performance & Delivery Efficiency • Last saved: Today at 1:13 PM

File Home Help Table tools Measure tools

Name: Measure Format: \$ % 00 Auto Data category: Uncategorized

Structure: 1 On-Time Trips = CALCULATE(COUNTROWS(Trip_Data), Trip_Data[Delivery_Status] = "On-Time")

Projects	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date
T01	V04	D01	DELHI	Pune	1173	108.42	On-Time	27 January 2023
T02	V06	D08	MUMBAI	Bangalore	1727	161.33	On-Time	21 February 2023
T03	V06	D08	MUMBAI	Pune	1459	154.70	On-Time	17 February 2023
T04	V04	D09	HYDERABAD	Pune	382	26.60	On-Time	18 February 2023
T05	V06	D08	PUNE	Mumbai	398	33.20	On-Time	15 February 2023
T06	V06	D07	CHENNAI	Mumbai	1275	85.04	Late	25 February 2023
T07	V07	D03	CHENNAI	Kolkata	752	58.08	On-Time	19 January 2023
T08	V02	D10	DELHI	Pune	74	5.24	On-Time	01 January 2023
T09	V02	D07	DELHI	Hyderabad	186	16.22	On-Time	23 February 2023
T010	V02	D02	BANGALORE	Hyderabad	1375	105.21	Late	02 February 2023
T011	V06	D03	KOLKATA	Hyderabad	419	31.17	On-Time	21 January 2023
T012	V06	D01	KOLKATA	Delhi	751	51.77	On-Time	15 February 2023
T013	V05	D04	KOLKATA	Chennai	1571	188.52	Late	02 February 2023
T014	V05	D05	HYDERABAD	Bangalore	1524	104.51	On-Time	16 February 2023
T015	V05	D06	KOLKATA	Mumbai	1956	179.88	On-Time	21 January 2023
T016	V05	D06	BANGALORE	Mumbai	858	92.70	Late	18 January 2023
T017	V07	D07	PUNE	Kolkata	1269	102.91	On-Time	16 January 2023
T018	V07	D10	PUNE	Delhi	1565	107.23	On-Time	12 February 2023
T019	V07	D10	HYDERABAD	Pune	1796	155.52	On-Time	28 February 2023
T020	V04	D02	MUMBAI	Bangalore	1640	148.87	On-Time	06 January 2023
T021	V07	D07	DELHI	Mumbai	446	31.98	Late	11 January 2023
T022	V07	D09	HYDERABAD	Delhi	140	11.44	On-Time	07 February 2023
T023	V06	D04	CHENNAI	Delhi	1729	107.55	On-Time	06 February 2023

Table: Trip_Data (50 rows) Column: Measure (0 distinct values)

❖ **DAX** played a significant role in enhancing my visualizations by enabling the creation of custom measures and calculations. The DAX formulas I used are listed below.

Formula: On-Time Trips = CALCULATE (COUNT ROWS(Trip_Data), Trip_Data[Delivery_Status] = "On-Time")

Formula: Total Trips = COUNTROWS (Trip_Data)

Formula: On-Time Delivery % = DIVIDE ([On-Time Trips], [Total Trips])

Formula: Total Distance = SUM(Trip_Data [Distance_km])

Formula: Total Fuel Consumed SUM(Trip_Data[Fuel_Consumed_L])

Formula: Fuel Efficiency DIVIDE ([Total Distance], [Total Fuel Consumed])

Formula: Total Maintenance Cost = SUM(Vehicle_Master[Maintenance_Cost])

Formula: Fuel Cost = [Total Fuel Consumed] * 100

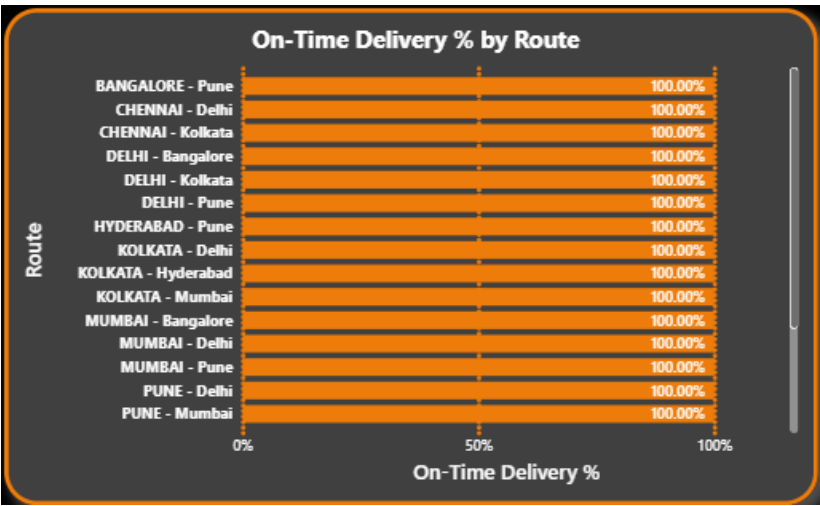
Formula: Cost per KM = DIVIDE([Fuel Cost] + [Total Maintenance Cost], [Total Distance])

DATA VISUALISATION



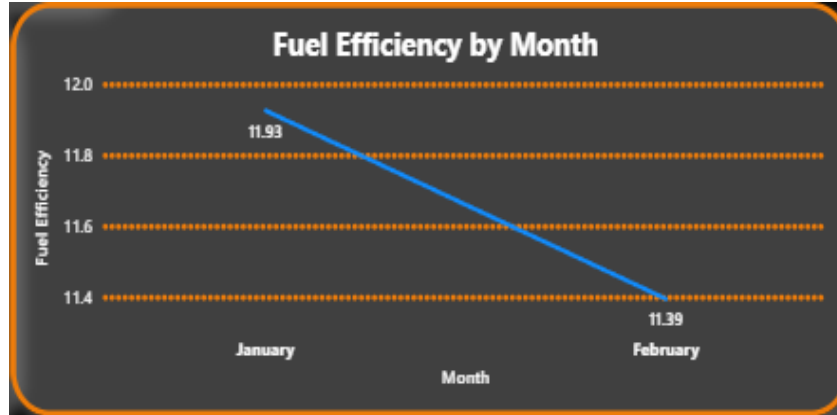
❖ I used KPI cards to visually present key metrics from the dataset, enabling quick and effective performance monitoring. These cards display important measures such as Total Trips, On-Time Delivery Percentage, Cost per Kilometer, and Fuel Efficiency. The On-Time Delivery Percentage was calculated to be 60%, and a screenshot of the KPI card is attached below for reference.

Clustered Bar Chart



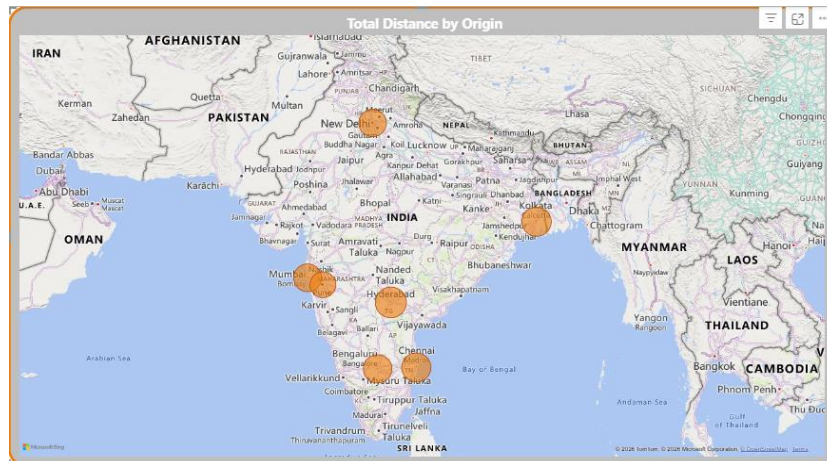
❖ Using a Clustered Bar Chart, we can analyze route performance and identify the routes with the highest number of on-time deliveries. This visualization helps pinpoint the exact routes that demonstrate the best delivery performance. According to the analysis, the Bangalore-to-Pune route ranks first in terms of on-time deliveries. The chart also highlights other top-performing routes, allowing for a comprehensive comparison of delivery efficiency across different routes.

Line Chart



- ❖ The month-wise fuel efficiency trend can be analyzed using the line chart. The chart shows that fuel efficiency was higher in January but declined in February, primarily due to the impact of the storm event. This visualization helps highlight fluctuations in fuel efficiency over time and the factors influencing operational performance.

Map Visual



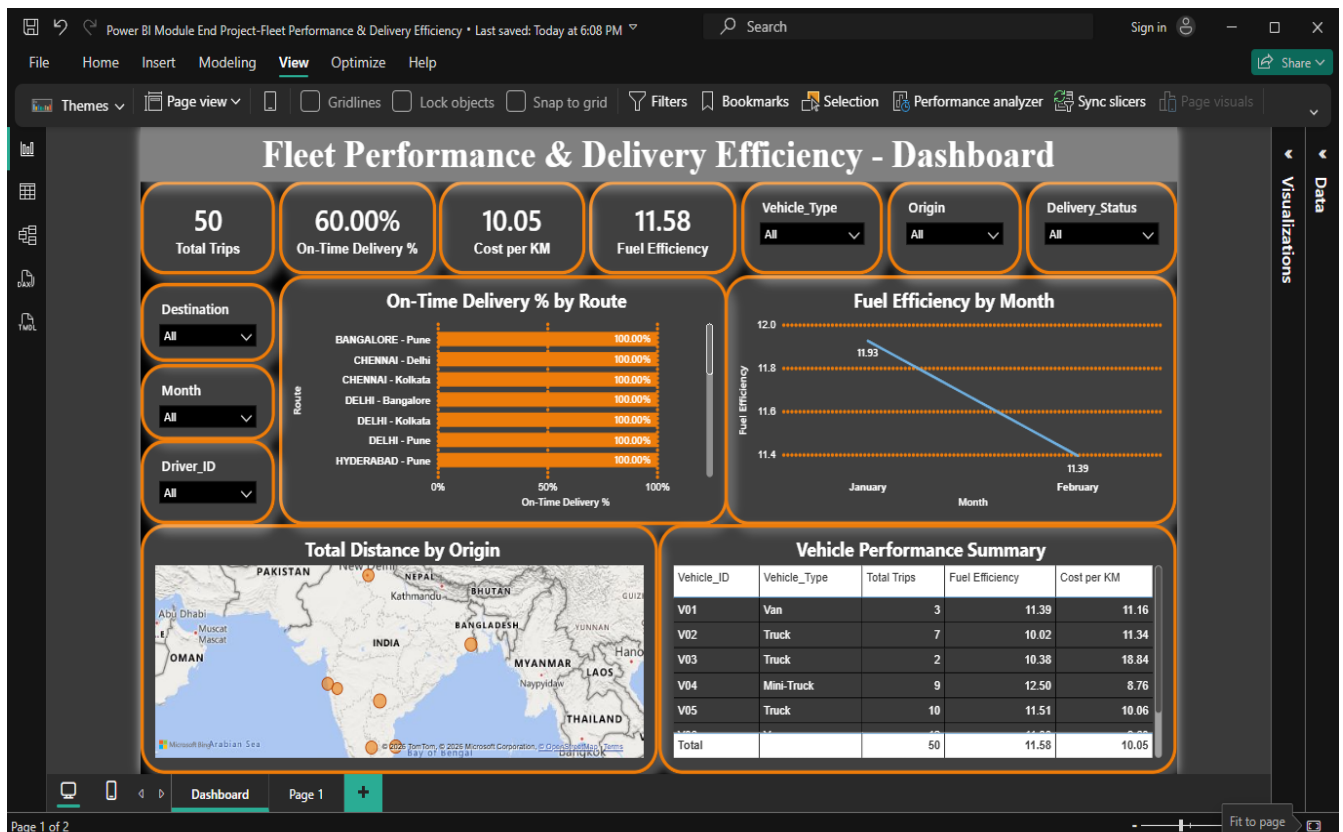
- ❖ By using the Map visual, we can analyze the total distance traveled from each origin location. This visualization provides a geographical view of the data, allowing us to examine trip distances across various cities, including Delhi, Pune, Chennai, and Bengaluru, and compare their transportation activity effectively.

Table

Vehicle_ID	Vehicle_Type	Total Trips	Fuel Efficiency	Cost per KM
V01	Van	3	11.39	11.16
V02	Truck	7	10.02	11.34
V03	Truck	2	10.38	18.84
V04	Mini-Truck	9	12.50	8.76
V05	Truck	10	11.51	10.06
Total		50	11.58	10.05

- ❖ I used a Table visual to analyze vehicle performance and included a screenshot below. It helps evaluate key metrics for each vehicle, such as Total Trips, Fuel Efficiency, and Cost per Kilometer.

DASHBOARD



Conclusion

- ❖ The Fleet Performance & Delivery Efficiency Dashboard successfully consolidates critical logistics data into a single, interactive Power BI report. With an overall on-time delivery rate of 60%, an average fuel efficiency of 11.58, and a cost per kilometer of 10.05 across 50 total trips, the dashboard offers a clear snapshot of current fleet operations. The vehicle performance table and route-level delivery charts highlight both strong performers and areas that require closer attention.

Recommendation

Investigate Low-Efficiency Vehicles: Vehicle V03 shows a high cost per KM (18.84) despite low trip count. A maintenance review is advised to determine if it should be retired or repaired.

Address the Fuel Efficiency Decline: Fuel efficiency dropped from 11.93 in January to 11.39 in February. Monthly tracking should continue to confirm whether this is a trend or an anomaly.

Expand Route Coverage: Most routes currently show 100% on-time delivery. As the fleet scales, the dashboard should be updated to capture more diverse route performance data.

Thank You !!!